Binary Search

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1 Binary Search

Binary Search

left $\leftarrow 0$
$n \leftarrow$ length of array A
right $\leftarrow n - 1$
while *left* $\leq$ *right* **do**
    mid $\leftarrow \frac{\text{left+right}}{2}$
    if *mid* $=$ *target* **then**
        return *mid*
    else if *mid* $\leq$ *target* **then**
        left $\leftarrow$ mid+1
    else
        right $\leftarrow$ mid-1
return *Not Found*

2 Lower-Upper Bound

Lower Bound

```
left ← 0
first_match ← -1
n ← length of array A
right ← n − 1
while left ≤ right do
|   mid ← (left+right)/2
|   if mid = target then
|   |   right = mid-1
|   |   first_match = mid
|   else if mid ≤ target then
|   |   left ← mid+1
|   else
|   |   right ← mid-1
return first_match
```

For upperbound, the right $=$ mid $-$ 1 condition in the if block will change to left $=$ mid $+$ 1, rest remains the same.